GearDesign Report

Cylindrical gear(pair)

작성일: 2025년 10월 22일

Contents

1. Summary

1.1. Gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.0000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	20.0000	
Manual normal backlash (mm)	j_bn	0.0000	
Number of teeth	z	29.0000	61.0000
Facewidth (mm)	b	40.0000	40.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	95.7760	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Gear material	-	DIN 18CrNiMo7 (σHlim = 430 Mpa)	DIN 18CrNiMo7 (σHlim = 430 Mpa)
Reference diameter (mm)	d	61.7220	129.8300
Tip diameter (mm)	d_a	65.722 (65.722 / 65.622)	133.830 (133.830 / 133.730)
Root diameter (mm)	d_f	56.722 (56.530 / 56.420)	124.830 (124.569 / 124.431)
Number of teeth spanned	k	4.0000	9.0000
Base tangent length (no backlash) (mm)	W_k	21.635 (21.569 / 21.531)	52.226 (52.137 / 52.090)
Effective Diameter of ball/pin (mm)	D_M	3.5000	3.5000
Diametral measurement over two balls (mm)	M_dk	66.620 (66.451 / 66.353)	134.829 (134.585 / 134.456)

1.2. Total safety factor

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]	Required S.F
Contact S.F	1.2513		1.2802		1.0000
Bending S.F	2.4772		2.5049		1.4000
Scuffing S.F (Flash Temp.)	3.4581				2.0000
Micropitting S.F	1.2626				1.0000

1.3. Total damage & life

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]
Contact Damage [%]	0.1000	0.0000	0.0000	0.0000
Bending Damage [%]	0.0000	0.0000	0.0000	0.0000
Contact Life [hr]	∞	∞	∞	∞
Bending Life [hr]	∞	∞	∞	∞
Total Req. Life [hr]	10000.0000			

1.4. Others

	Gear1 - Gear2
Transverse Contact Ratio	1.4993
Overlap Ratio	2.1774
Total Contact Ratio	3.6767
Worst Pitch Line Velocity [m/s]	32.3177
Total Efficiency [%]	98.6415
Worst Mesh Loss [W]	560.2872
Worst Churning Loss [W]	1477.4136
Worst Peak-to-Peak T.E. [μm]	0.3700
Gear Total Mass [kg]	1.4887
Hunting tooth	Complete

1.5. Summary - Load cases

	Gear	Time [Hr]	Temperautre [ºC]	Power [kW]	Speed [r/min]	Torque [N.m]	Contact SF (Left)	Bending SF (Left)	Scuffing SF (Left)	Contact SF (Right)	Bending SF (Right)	Scuffing SF (Right)	K_Hβ (Left)	K_Hβ (Right)	Mesh Loss (Left) [kW]	Mesh Loss (Right) [kW]	Churning Loss [kW]	PPTE (Left) [μm]	PPTE (Right) [μm]
LC	Gear1	10000.0000	80.0000	150.0000	10000.0000	143.2390	1.2513	2.4772	3.4581				1.3000		0.5603		0.2139	0.3700	
1	Gear2			-150.0000	-4754.0980	301.2970	1.2802	2.5049									1.2636		

2. Detail geometry

2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.0000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	20.0000	
Manual normal backlash (mm)	j_bn	0.0000	
Number of teeth	z	29.0000	61.0000
Facewidth (mm)	b	40.0000	40.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	95.7760	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Addendum coefficient	h*_aP	1.0000	1.0000
Dedendum coefficient	h*_fP	1.2500	1.2500
Root radius coefficient	ρ*_fP	0.3800	0.3800
Tip form height coefficient	h*_FaP	0.9500	0.9500
Ramp angle (deg)	α_KP	50.0000	50.0000
Protuberance height coefficient	h*_prP	0.0000	0.0000
Protuberance angle (deg)	α_prP	0.0000	0.0000
Tip alteration (mm)	k.m_n	0.0000	0.0000
Tip radius coefficient (for topping)	ρ*_aP	0.0000	0.0000
Gear material	-	DIN 18CrNiMo7 (σHlim = 430 Mpa)	DIN 18CrNiMo7 (σHlim = 430 Mpa)
Young's modulus (N/mm^2)	E	2.0600×10^5	2.0600×10^5
Poisson's ratio	ν	0.3000	0.3000
Density (kg/m^3)	ρ	7830.0000	7830.0000
Coeff. of thermal expansion (10^-8/ºC)	α_coeff	11.5000	11.5000
Surface hardness	HRC	61.0000	61.0000
Core hardness	HBW	325.0000	325.0000
Tensile strength (N/mm^2)	σ_B	1200.0000	1200.0000
Yield strength (N/mm^2)	σ_s	850.0000	850.0000
Specific heat capacity (J/kg.K)	c_M	485.0000	485.0000
Specific heat conductivity (W/m.K)	λ_M	50.0000	50.0000
Mean roughness height, root (um)	R_ZF	20.0000	20.0000
Mean roughness height, flank (um)	R_ZH	4.8000	4.8000
Roughness average value, root (um)	R_AF	3.0000	3.0000
Roughness average value, flank (um)	R_AH	0.6000	0.6000

	Symbol	Pair1 Gear1	Pair1 Gear2
Fatigue strength for bending stress (N/mm^2)	σ_Flim	430.0000	430.0000
Fatigue strength for contact stress (N/mm^2)	σ_Hlim	1500.0000	1500.0000

2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Gear ratio	GR	2.1034	
Transverse module (mm)	m_t	2.1284	
Working normal pressure angle (deg)	α_wn	20.0000	
Transverse pressure angle (deg)	α_t	21.1728	
Working transverse pressure angle (deg)	α_wt	21.1728	
Working transverse pressure angle, e/i (deg)	α_wt, e/i	21.1898 / 21.1558	
Base helix angle (deg)	β_b	18.7472	
Helix angle at operating pitch circle (deg)	β_w	20.0000	
Radial backlash without tolerance (mm)	j_r0	0.0000	
Circumferential backlash without tolerance (mm)	j_t0	0.0000	
Min/max circumferential backlash by flank tolerance for each gear (mm)	j_t, e/i	0.0745 / 0.1171	0.1011 / 0.1543
Min/max normal backlash by flank tolerance for each gear (mm)	j_n, e/i	0.0658 / 0.1034	0.0893 / 0.1363
Circumferential backlash, e/i (mm)	j_tw, e/i	0.1671 / 0.2799	
Normal backlash, e/i (mm)	j_nw, e/i	0.1475 / 0.2471	
Angluar backlash, min (deg)	j_θ, min	0.3102	0.1475
Angluar backlash, max (deg)	j_θ, max	0.5196	0.2470
Effective facewidth (mm)	b_eff	40.0000	
Sum of the profile shift coefficients	Sum_x	0.0000	
Generating profile shift coefficient	x_E, e/i	-0.0481 / -0.0756	-0.0653 / -0.0996
Prevent undercut profile shift coefficient	x_Emin	-1.0130	-3.2342
Max. profile shift coeff. (min top land)	x_Emax	1.4605	2.3748
Reference center distance, x1=x2=0 (mm)	a_d	95.7760	
Working center distance (mm)	a_w	95.7760	
Reference diameter (mm)	d	61.7223	129.8297
Operating pitch diameter (mm)	d_w	61.7223	129.8297
Operating pitch diameter with tolerance (mm)	d_w, e/i	61.7294 / 61.7152	129.8446 / 129.8148
27		65.7223	133.8297
Tip diameter with tolerance (mm)	d_a, e/i	65.7223 / 65.6223	133.8297 / 133.7297

	Symbol	Pair1 Gear1	Pair1 Gear2
Root diamter (mm)	d_f	56.7223	124.8297
Root diameter with tolerance (mm)	d_f, e/i	56.5300 / 56.4201	124.5687 / 124.4313
Tip form diameter (mm)	d_Fa	65.5223	133.6297
Tip form diameter with tolerance (mm)	d_Fa, e/i	65.3108 / 65.1904	133.3552 / 133.2111
Root form diamter (mm)	d_Ff	58.6389	126.2529
Root form diameter with tolerance (mm)	d_Ff, e/i	58.5394 / 58.4846	126.0498 / 125.9444
Active tip diameter (EAP) (mm)	d_Na	65.5223	133.6297
Active tip diameter with tolerance (mm)	d_Na, e/i	65.3108 / 65.1904	133.3552 / 133.2111
Active root diameter (SAP) (mm)	d_Nf	58.9223	126.8510
Active root diameter with tolerance (mm)	d_Nf, e/i	59.1577 / 59.0516	127.0804 / 126.9662
Mean diameter for mass calculation (mm)	d_m	61.0737	129.1398
Gear mass (kg)	m	0.4394	1.0493
Gear inertia (kg.m^2)	J	0.0000	0.0000
Nominal tip clearance (mm)	c	0.5000	0.5000
Effective tip clearance, e/i (mm)	c, e/i	0.7602 / 0.6195	0.7121 / 0.5852
Min. margin of involute contact, (d_Nf_i - d_Ff_e) / 2 (mm)	c_NfFf	0.2561	0.4582

2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Tooth depth (mm)	h	4.5000	4.5000
Tooth depth, e/i (mm)	h, e/i	4.6511 / 4.5462	4.6992 / 4.5805
Working depth (mm)	h_w	4.0000	
Addendum (mm)	h_a	2.0000	2.0000
Addendum, e/i (mm)	h_a, e/i	2.0000 / 1.9500	2.0000 / 1.9500
Dedendum (mm)	h_f	2.5000	2.5000
Dedendum, e/i (mm)	h_f, e/i	2.5962 / 2.6511	2.6305 / 2.6992
Number of virtual gear teeth	z_n	34.4161	72.3926
Tooth thickness coeff.	s_nc	1.5708	1.5708
Transverse tooth thickness (mm)	s_t	3.3432	3.3432
Normal tooth thickness (mm)	s_n	3.1416	3.1416
Normal tooth thickness at d_a (mm)	s_an	1.5016	1.5910
Normal tooth thickness at d_a, e/i (mm)	s_an, e/i	1.4770 / 1.3854	1.5364 / 1.4421
Normal tooth thickness at d_Fa (mm)	s_Fan	1.5997	1.6767
Normal tooth thickness at d_Fa, e/i (mm)	s_Fan, e/i	1.6854 / 1.5860	1.7563 / 1.6446
Normal space width at d_f (mm)	e_fn	0.0000	1.5382
Normal space width at d_f, e/i (mm)	e_fn, e/i	0.0000 / 0.0000	1.5662 / 1.5817

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal pitch (mm)	p_n	6.2832	
Transverse pitch (mm)	p_t	6.6864	
Normal base pitch (mm)	p_bn	5.9043	
Transverse base pitch (mm)	p_bt	6.2351	
Axial pitch (mm)	p_x	18.3708	
Lead (mm)	p_z	532.7533	1120.6191

2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Length of the path of contact (A to E) (mm)	g_α	9.3483	
Length of path of contact, e/i (mm)	g_α, e/i	8.8305 / 8.4695	
Length of approach POC of pinion (A to C) (mm)	g_f1	4.8383	
Length of approach POC of pinion, e/i (mm)	g_f1, e/i	4.5331 / 4.4919	
Length of recess POC of pinion (C to E) (mm)	g_a1	4.5100	
Length of recess POC of pinion, e/i (mm)	g_a1, e/i	4.2974 / 4.2777	
Length T1 - A (mm)	T1A	6.3082	
T1A tolerance (mm)	e/m/i	6.8367 / 6.7201 / 6.6035	
Length T1 - B (mm)	T1B	9.4215	
T1B tolerance (mm)	e/m/i	9.1990 / 9.1351 / 9.0712	
Length T1 - C (mm)	T1C	11.1465	
T1C tolerance (mm)	e/m/i	11.1563 / 11.1465 / 11.1367	
Length T1 - D (mm)	T1D	12.5433	
T1D tolerance (mm)	e/m/i	13.0718 / 12.9552 / 12.8386	
Length T1 - E (mm)	T1E	15.6565	
T1E tolerance (mm)	e/m/i	15.4340 / 15.3701 / 15.3062	
Length T2 - A (mm)	T2A	28.2844	
T2A tolerance (mm)	e/m/i	27.9586 / 27.8725 / 27.7863	
Length T2 - B (mm)	T2B	25.1711	
T2B tolerance (mm)	e/m/i	25.5519 / 25.4575 / 25.3632	
Length T2 - C (mm)	T2C	23.4461	
T2C tolerance (mm)	e/m/i	23.4667 / 23.4461 / 23.4255	
Length T2 - D (mm)	T2D	22.0493	
T2D tolerance (mm)	e/m/i	21.7235 / 21.6374 / 21.5513	
Length T2 - E (mm)	T2E	18.9361	
T2E tolerance (mm)	e/m/i	19.3168 / 19.2225 / 19.1281	
Length T1 - T2 (mm)	T1T2	34.5926	

	Symbol	Pair1 Gear1	Pair1 Gear2
T1T2 tolerance (mm)	e/m/i	34.6231 / 34.5926 / 34.5621	
Diameter of single contact point A (mm)	d_A	58.9223	133.6297
Diameter of single contact point B (mm)	d_B	60.5617	131.1153
Diameter of single contact point C (mm)	d_C	61.7223	129.8297
Diameter of single contact point D (mm)	d_D	62.7854	128.8470
Diameter of single contact point E (mm)	d_E	65.5223	126.8510
Transverse contact ratio	ε_α	1.4993	
Overlap ratio	ε_β	2.1774	
Total contact ratio	ε_Y	3.6767	
Total contact ratio with tolerance	e/m/i	3.594 / 3.565 / 3.536	
Specific sliding at Tip	ζ_a	0.4250	0.5309
Specific sliding at Root	ζ_f	-1.1316	-0.7392

2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Quality Standard	-	ISO 1328: 2013	ISO 1328: 2013
Single pitch deviation (μm)	f_pt	8.5000	8.5000
Base circle pitch deviation (μm)	f_pb	7.9000	7.9000
Sector pitch deviation over k/8 pitches (μm)	F_pk/8T	18.0000	19.0000
Profile form deviation (μm)	f_fα	8.5000	8.5000
Profile slope deviation (μm)	f_Hα	7.0000	7.0000
Total profile deviation (μm)	F_α	11.0000	11.0000
Helix form deviation (μm)	f_fβ	10.0000	11.0000
Helix slope deviation (μm)	f_Hβ	9.5000	9.5000
Total helix deviation (μm)	F_β	14.0000	15.0000
Cumulative pith deviation, total (μm)	F_p	25.0000	28.0000
Runout (μm)	F_r	22.0000	25.0000
Single flank composite, tooth-to-tooth (μm)	f_isT	8.0000	8.0000
Single flank composite, total (μm)	F_isT	33.0000	36.0000
Tooth-to-tooth radial composite deviation (μm)	f_idT	9.5000	9.5000
Total radial composite deviation (μm)	F_idT	31.0000	37.0000

2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Number of teeth spanned	k	4.0000	9.0000
Base tangent length (mm)	W_k	21.6347	52.2261
Base tangent length with tolerance (mm)	W_k, e/i	21.5689 / 21.5313	52.1368 / 52.0899
Tolerance of W_k (mm)	ΔW_k, e/i	-0.0658 / -0.1034	-0.0893 / -0.1363
Diameter of contact point (mm)	d_M	61.0664	130.7369
Theoretical diameter of ball/pin (mm)	D_M.theory	3.4000	3.3694
Effective Diameter of ball/pin (mm)	D_M.eff	3.5000	3.5000
Radial single ball measurement (mm)	M_rk	33.3563	67.4363
Radial single ball measurement with tolerance (mm)	M_rk, e/i	33.2717 / 33.2229	67.3143 / 67.2495
Diametral two ball measurement (mm)	M_dk	66.6199	134.8291
Diametral two ball measurement with tolerance (mm)	M_dk, e/i	66.4510 / 66.3535	134.5851 / 134.4556
Tolerance of M_dk (mm)	ΔM_dk	-0.1689 / -0.2664	-0.2440 / -0.3735

3. Detail rating

3.1. General

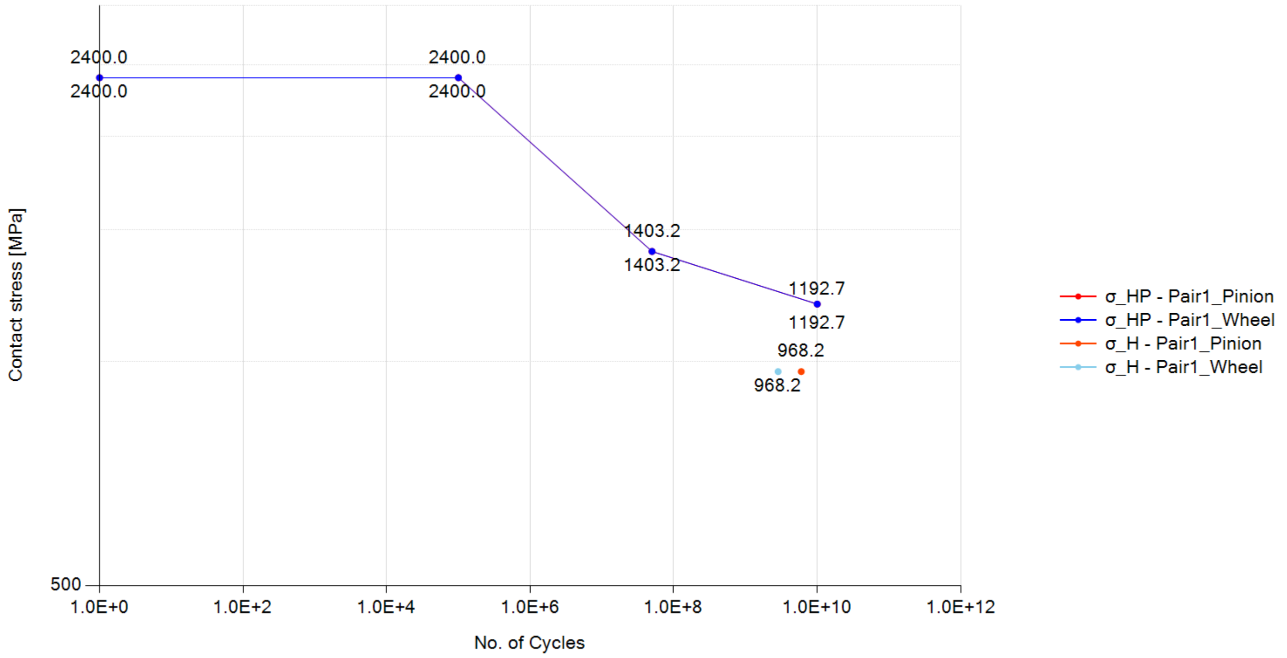
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rotational speed (rpm)	ω	10000.0000	4754.0980
Nominal tangential load at d (N)	F_t	4641.4160	
Nominal axial load at d (N)	F_a	1689.3370	
Nominal radial load at d (N)	F_r	1797.7550	
Nominal normal load at d (N)	F_n	5256.2840	
Nominal tangential load per unit facewidth (N/mm)	w	116.0350	
Circumferential speed at d (m/s)	v	32.3180	
Sliding velocity at Tip dia. (m/s)	v_g	6.9680	7.4750
Nominal tangential load at d_w (N)	F_tw	4641.4160	
Nominal axial load at d_w (N)	F_aw	1689.3370	
Nominal radial load at d_w (N)	F_rw	1797.7550	
Circumferential speed at d_w (m/s)	v_w	32.3180	
Running-in value (um)	y_p	0.5930	
Running-in value (um)	y_f	0.6370	
Effective single pitch & profile deviation (um)	f_pb_eff	7.3080	
Effective single pitch & profile deviation (um)	f_fa_eff	7.8620	
Correction factor	C_M	0.8000	
Gear blank factor	C_R	0.8980	
Basic rack factor	C_B	0.9750	
Material factor	E/E_st	1.0000	
Theoretical single stiffness (N/mm/um)	c'_th	18.0800	
Single tooth stiffness (N/mm/um)	c'	11.9000	
Mesh stiffness (for Kv, KHα, KFα), (N/mm/um)	c_yα	16.3560	
Mesh stiffness (for KHβ, KFβ), (N/mm/um)	c_yβ	13.9030	
Mean diameter for K_v (mm)	d_m (K_V)	61.2220	129.3300
Moment of inertia for K_v (kg.mm^2)	J (K_V)	376.0600	4889.0100
Moment of gear inertia per unit faccewidth (kg.mm)	J*	9.4020	122.2250
Relative gear mass per unit facewidth (kg/mm)	m*	0.0114	0.0334
Relative gear mass per unit facewidth (kg/mm)	m_red	0.0085	
Resonance running speed (rpm)	n_E1	14470.4370	
Resonance ratio	N	0.6910	
Resonance ratio in main resonance ratio	N_s	0.8500	
Resonance range	-	subcritical range	
Dynamic factor	K_V	1.2390	

	Symbol	Pair1 Gear1	Pair1 Gear2
Face load factor - flank	K_Hβ	1.3000	
Face load factor - root	K_Fβ	1.2610	
Transverse load factor - flank	K_Hα	1.1670	
Transverse load factor - root	K_Fα	1.1670	
Number of load cycles	N_L	6.0000×10^9	2.8520×10^9

3.2. Pitting

S-N curve (Pitting), Load case 1, Meshing pair: Gear1 - Gear2

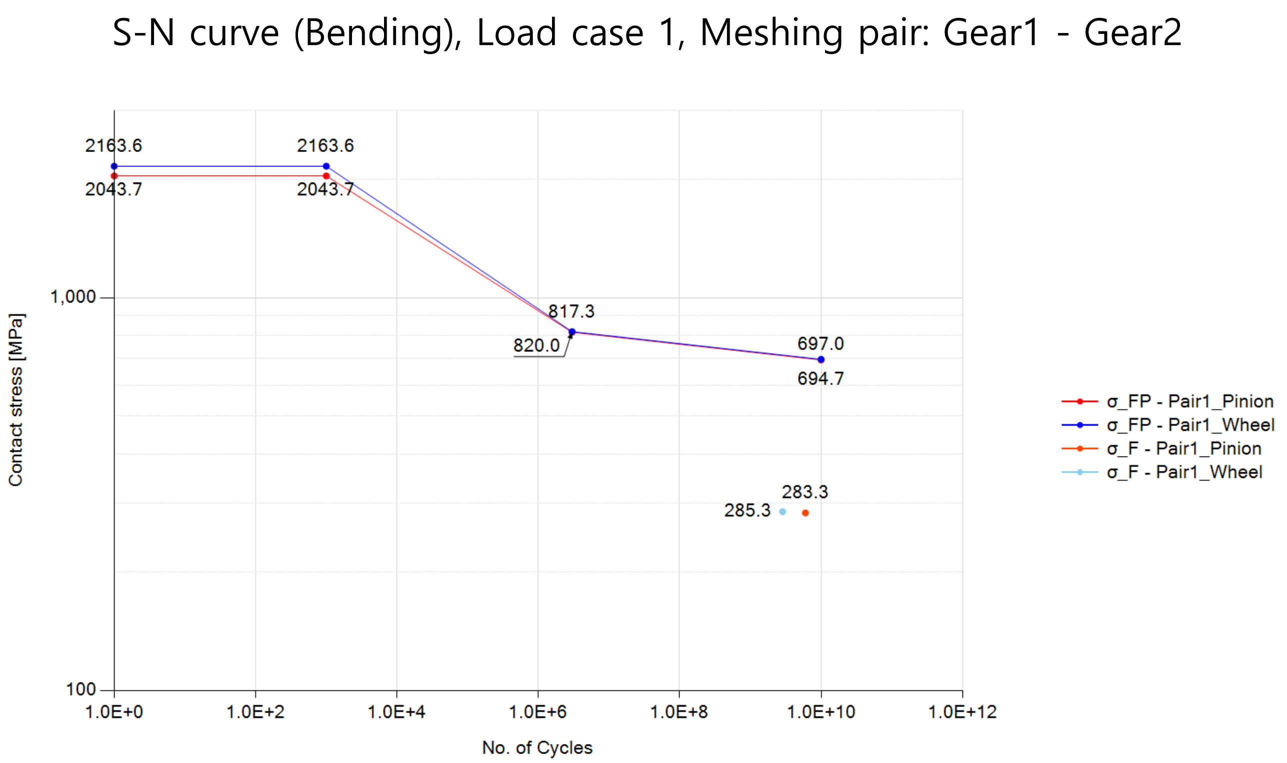


Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-2:2019	
Contact ratio factor	Z_ε	0.8170	
Zone factor	Z_H	2.3710	
Elasticity factor	Z_E	189.8120	
Young's modulus for tooth flank (N/mm^2)	E	2.0600×10^5	2.0600×10^5
Helix angle factor	Z_β	1.0320	
Single pair tooth contact factor (pinion)	Z_B	1.0000	
Single pair tooth contact factor (wheel)	Z_D	1.0000	
Nominal contact stress at the pitch point (N/mm^2)	σ_H0	631.5470	
Contact stress (N/mm^2)	σ_H	968.2020	968.2020
Life factor	Z_NT	0.8630	0.8830

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubricant factor	Z_L	0.9460	0.9460
Velocity factor	Z_V	1.0350	1.0350
Roughness factor	Z_R	0.9560	0.9560
Work hardening factor	Z_W	1.0000	1.0000
Size factor	Z_X	1.0000	1.0000
Permissible contact stress - static (N/mm^2)	σ_HP_static	2400.0000	2400.0000
Permissible contact stress - mid point (N/mm^2)	σ_HP_mid	0.0000	0.0000
Permissible contact stress - reference (N/mm^2)	σ_HP_ref	1403.1580	1403.1580
Permissible contact stress - long (N/mm^2)	σ_HP_long	1192.6840	1192.6840
Permissible contact stress (N/mm^2)	σ_HP	1211.5190	1239.4700
Safety factor for Contact	S_H	1.2510	1.2800
Contact damage (%)	-	0.0670	0.0320
Cycle to failure	-	8.9610×10^12	8.9610×10^12

3.3. Bending



Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-3:2019	
Tooth form factor	Y_F	1.1300	1.0660
Stress correction factor	Y_S	1.8880	2.0140

	Symbol	Pair1 Gear1	Pair1 Gear2
Working angle (deg)	α_Fen	18.9010	19.7880
Bending moment arm (mm)	h_Fe	2.0690	2.2240
Tooth thickness at root (mm)	s_Fn	4.1370	4.4030
Tooth root radius (mm)	ρ_F	1.1280	1.0220
Helix angle factor	Y_β	1.0040	
Deep tooth factor	Y_DT	1.0000	
Rim thickness factor	Y_B	1.0000	1.0000
Nominal tooth root stress (N/mm^2)	σ_F0	124.2300	125.1160
Tooth root stress (N/mm^2)	σ_F	283.3280	285.3480
Life factor	Y_NT	0.8590	0.8720
Relative notch sensitive factor	Y_δrelT	0.9930	0.9970
Surface factor	Y_RrelT	0.9570	0.9570
Size factor	Y_X	1.0000	1.0000
Stress correction factor	Y_ST	2.0000	
Mean stress influence factor	Y_M	1.0000	1.0000
Technical factor	Y_T	1.0000	1.0000
Permissible tooth root stress - static (N/mm^2)	σ_FP_static	2043.6550	2163.5830
Permissible tooth root stress - reference (N/mm^2)	σ_FP_ref	817.2990	820.0330
Permissible tooth root stress - long (N/mm^2)	σ_FP_long	694.7040	697.0280
Permissible tooth root stress (N/mm^2)	σ_FP	701.8500	714.7680
Safety factor for Bending	S_F	2.4770	2.5050
Bending damage (%)	-	0.0000	0.0000
Cycle to failure	-	1.4050×10^22	1.1650×10^22

3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubrication type	-	Spray lubrication	
Lubricant temperature (deg)	θ_oil	80.0000	
Lubricant viscosity (mPa.s)	η_oil	13.1510	
Load stage according to FZG A/8, 3/90	S_FZG	12.0000	
Gear density (kg/m^3)	ρ_M	7830.0000	7830.0000
Gear specific heat (J/kg.K)	c_M	485.0000	485.0000
Gear heat conductivity (N/s.K)	λ_M	50.0000	50.0000
Gear roughness (μm)	R_a	0.6000	0.6000
Number of mating gears	n_p	1.0000	
Effective facewidth (mm)	b_eff	34.0000	

	Symbol	Pair1 Gear1	Pair1 Gear2
Relevant tip relief (μm) (Min. value = 2 μm)	C_a	2.0000	2.0000
Transverse unit load (N/mm)	w_Bt	272.7170	
Lubrication system factor	X_S	1.2000	
Multiple meshing factor	X_mp	1.0000	
Structural factor	X_W	1.0000	1.0000
Structural factor (total)	X_Wtot	1.0000	
Thermal contact factor	B_M	13.7800	13.7800
Multiple path factor	K_mp	1.0000	
Optimal tip relief (μm)	C_eff	8.8680	
Angle factorr	X_αβ	0.9690	
Lubricant factor	X_L	0.8790	
Roughness factor	X_R	0.7750	
Mean coefficient of friction	μ_m	0.0450	
Maximum flash temperature (deg)	θ_fl_max	68.5590	
Mean flash temperature (deg)	θ_fl_mean	24.4850	
Total mass temperature (deg)	θ_M	93.8100	
Scuffing temperature (deg)	θ_S	364.8370	
Maximum contact temperature (deg)	θ_Bmax	162.3680	
Safety factor for flash temperature	S_B	3.4580	

3.5. Micro-pitting

Load case 1

	Symbol	Gear1 - Gear2
minimum lubricant film thickness (μm)	h_Ymin	0.1599
minimum specific lubricant film thickness in the contact area	λ_GFmin	0.2665
permissible specific lubricant film thickness	λ_GFP	0.2111
safety factor against micropitting	S_λ	1.2626
transverse tangential load at reference cylinder per mesh (N)	F_t	4641.4160
transverse load in plane of action (N)	F_bt	4977.4150
reduced modulus of elasticity (N/mm^2)	E_r	2.2637×10^5
thermal contact coefficient of pinion (N/(ms^0.5*K))	B_M1	13779.6040
thermal contact coefficient of wheel (N/(ms^0.5*K))	B_M2	13779.6040
failure load stage according to FVA 54/7 (90º)	-	0.0000
effective arithmetic mean roughness value (μm)	R_a	0.6000
roughness factor	X_R	1.1520
lubricant factor	X_L	1.0000
lubrication factor	X_S	1.2000

	Symbol	Gear1 - Gear2
elasticity factor (N/mm^2)^0.5	Z_E	189.8120
helical load factor	K_Bγ	1.3000
load losses factor	H_v	0.1060
mean coefficient of friction	μ_m	0.0530
profile position at min. lubricant film thickness	-	0.0000
tip relief factor	X_Ca	1.0000
load sharing factor	X_A	0.8670
buttressing factor	X_but,A	1.3000
effective tip relief (μm)	C_eff	0.0000
Applied tip relief in calculation (μm)	C_a	0.0000
sum of tangential velocities at point A (m/s)	v_Σ,A	20.6870
sliding velocity at point A (m/s)	v_g,A	-7.4750
tangential velocity on pinion at point A (m/s)	v_r1,A	6.6060
tangential velocity on wheel at point A (m/s)	v_r2,A	14.0810
oil inlet/sump temperature (ºC)	θ_oil	80.0000
bluk temperature (ºC)	θ_M	97.1010
flash temperature at point A (ºC)	θ_fl,A	81.9560
contact temperature at point A (ºC)	θ_B,A	179.0570
pressure-viscosity coefficient at 38 ºC (m^2/N)	α_38	1.8390×10^-8
pressure-viscosity coefficient at bulk temperature (m^2/N)	α_θM	1.3520×10^-8
pressure-viscosity coefficient at point A contact temperature (m^2/N)	α_θB,A	8.8700×10^-9
kinematic viscosity at 40 ºC (mm^2/s)	ν_40	68.0000
kinematic viscosity at 100 ºC (mm^2/s)	ν_100	8.8000
kinematic viscosity at point A contact temperature (mm^2/s)	ν_θB,A	2.3890
dynamic viscosity at 38 ºC (Ns/m^2)	η_38	0.0650
dynamic viscosity at bulk temperature (Ns/m^2)	η_θM	0.0080
dynamic viscosity at oil inlet/sump temperature (Ns/m^2)	η_θoil	0.0120
dynamic viscosity at point A contact temperature (Ns/m^2)	η_θB,A	0.0020
normal radius of relative curvature at point A (mm)	ρ_n,A	5.4470
material parameter	G_M	3060.2750
local velocity parameter at point A	U_A	6.5450×10^-11
local load parameter at point A	W_A	0.0002
local sliding parameter at point A	S_GF,A	0.1550
local nominal Hertzian contact stress at point A (Method B) (N/mm^2)	p_H,A	844.7880
local Hertzian contact stress at point A including the load factors K (Method B) (N/mm^2)	p_dyn,A	1295.1140

4. Power loss

4.1. Mesh loss

Load case 1

	Symbol	Gear1 - Gear2
Rating Method	-	ISO/TR 14179-2:2001
Input Power (kW)	P	150.0000
Gear ratio	u	2.1034
Tangential speed at working pitch dia. (m/s)	v_wt	32.3177
Tangential force at pitch dia. (N)	F_t	4641.4156
Tooth normal force, transverse section (N)	F_bt	4977.4152
Arithmetic average roughness of pinion	Ra1	0.6000
Arithmetic average roughness of wheel	Ra2	0.6000
Roughness factor	X_R	0.6000
Sum velocity (m/s)	v_Σ	23.3452
Equivalent radius of curvature (mm)	ρ	7.9781
Kinematic viscosity of oil at operating temp. (cSt)	η_oil	14.8602
Lubricant type	-	Mineral oil base
Lubricant factor	X_L	1.0000
distributed force on tooth surface(N/m)	F/b	150.0000
Mean coefficient of friction of the gear mesh	μ_mz	0.0353
Tooth loss factor	H_v	0.1057
Mesh power loss (W)	P_VZP	560.2872
Mesh efficiency (%)	η_z	99.6265

4.2. Churning loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	Changenet (2011)	Changenet (2011)
Rotating speed (rpm)	n	10000.0000	4754.0984
Lubricant	-	Oil: ISO-VG 68	Oil: ISO-VG 68
Gear immerged depth (mm)	h_e	30.8612	64.9148
Oil density (kg/m^3)	ρ	885.0000	885.0000
Operating Viscocity (cSt)	ν	14.8602	14.8602
dynamic viscosity (kg/m.s)	μ	0.0000	0.0000
Reynolds number	Re	86991.3862	86991.3868
Froude number	Fr	3449.8516	1640.0934
Centrifugal acceleration (m/s^2)	γ	14821.4704	4292.1171

	Symbol	Gear1	Gear2
Tip diameter of gear (mm)	d_a	65.7223	133.8297
Pitch diameter of gear (mm)	d_p	61.7223	129.8297
Facewidth (mm)	b	40.0000	40.0000
Dimensionless drag torque	C_m	0.0011	0.0025
Immersion area of the gear (m^2)	S_m	0.0128	0.0338
Total oil churning loss (W)	P	213.8512	1263.5624
Total oil churning torque (N.m)	T	0.2042	2.5380

4.3. Windage loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	None	None
Windage loss (W)	P_WL	0.0000	0.0000
Windage torque (N.m)	T_WL	0.0000	0.0000